Silicon Photonics for AI - Balancing Cost, Power, and Reliability

来源: NADDOD Blog (naddod.com) | 作者: Holly | 发布日期: 2024年9月6日 | 分类: Optical Transceivers
原文: https://www.naddod.com/blog/silicon-photonics-for-ai-balancing-cost-power-and-reliability

存档说明：该页面因厂商网站地区重定向无法直接抓取，本 PDF 内容根据搜索引擎索引的文章完整段落重建，覆盖原文主要章节（引言、低成本、低功耗、高可靠性），个别过渡语句可能与原文略有出入。

Silicon photonics technology in AI scenarios prioritizes three core demands: low cost, low power consumption, and high reliability, aligning with NVIDIA's requirements.

Low Cost

Aiming for a cost of $0.25 per Gbps, which is about a quarter of current optical module costs. The goal is to afford a high-capacity, highly integrated 3.2T silicon photonics module at the price of today's 800G module. Broadcom's 2023 release of a 6.4T CPO silicon photonics module shows promising advancements in reducing costs as the industry matures.

Low Power Consumption

Achieving low power consumption remains a significant technical challenge. For instance, to meet the demanding power efficiency goals for AI silicon photonics modules, current 800G DR8 modules consume 13W, equating to 16 picojoules per bit (pJ/bit). Reducing this to 1.5 pJ/bit requires a 90% reduction in power consumption. Increasing bandwidth is one approach, but more significantly, CPO (Co-Packaged Optics) packaging can eliminate the need for DSPs, cutting power use in half without sacrificing RF performance, unlike LPO (Linear-drive Pluggable Optics). Replacing MZ modulators with micro-ring modulators can further reduce energy consumption.

In 2022, Broadcom's analysis of power distribution in switches showed that lasers and CPO components (modulators, detectors, DRV, TIA) share power consumption evenly, with their CPO based on cascaded MZ modulators achieving 5.5 pJ/bit. MZ modulators have VpiL of about 1-2 V・cm, while micro-ring modulators reduce this to 0.5 V・cm, enhancing efficiency with a smaller footprint. In 2023, IMEC conducted a detailed analysis of the power consumption for systems based on micro-ring modulators (4 pJ/bit). Ranovus uses micro-ring modulators. Intel also employs micro-ring modulators, and in 2023, Intel's CPO with integrated lasers demonstrated power consumption of 3 pJ/bit.

Why does micro-ring need thermal compensation? Micro-rings require thermal compensation due to resonance wavelength drift under varying bias voltages and temperatures, with drift coefficients around 0.03 nm/V for voltage and 0.1 nm/°C for temperature, necessitating precise wavelength stabilization when interfacing lasers with micro-ring modulators. The wavelength stability of micro-ring modulators can be managed by either thermal tuning or electrical tuning. NVIDIA has specified a thermal tuning precision of 0.4°C. However, thermal tuning consumes significant power, while electrical tuning offers weaker performance. HP's approach eliminates thermal tuning but uses PN heterojunctions of silicon and GaAs instead of pure silicon, which impacts reliability. This results in trade-offs between performance, power consumption, and reliability in micro-ring designs. Regarding thermal tuning, challenges include crosstalk and mutual thermal effects among cascaded micro-rings. To maximize wavelength alignment between micro-ring resonators and lasers, integrating lasers made of InP with silicon modulators is beneficial as their temperature drift coefficients are similar, reducing compensation difficulties. However, integrating lasers also

introduces reliability risks, as lasers are among the most failure-prone components in optical modules.

High Reliability

Regarding reliability, conventional optical modules generally require a laser FIT (Failure In Time rate) of 125, meaning a 5-year lifespan with a failure rate of 0.5%. Silicon photonics modules now target 100 FIT, with a 5-year failure rate of 0.4%, which appears acceptable. However, silicon photonics involve multiple integrated channels - like Broadcom's 6.4T CPO with 64 modulators and Intel's CPO with 96 micro-ring modulators per chip. Each modulator requires a laser, and with a laser FIT of 100, a 10-laser array could escalate the failure rate to 1000 FIT, or 4%.

A key challenge is managing the FIT rate as it scales with laser array size. HP addresses this by using optical frequency combs with multi-wavelength lasers, where a single laser supports multiple micro-ring modulators, balancing low power consumption and high reliability. Intel initially paired one laser with each micro-ring modulator but began developing multi-wavelength lasers in 2023, reducing the number of lasers and thereby lowering the failure rate. Another approach, similar to Finisar's, involves adding backup lasers. Finisar developed a VCSEL-based multimode CPO for IBM, doubling the laser count and reducing the failure rate by three orders of magnitude. Intel also evaluated the reliability of backup lasers between 2020 and 2021, showing that backup lasers can significantly reduce failure rates.